

**Laplace transform of some elementary functions:**

Continuation...

$f(t)$	$F(s)$
6. $e^{\pm at} \sin kt$	$\frac{k}{(s \mp a)^2 + k^2}$
7. $e^{\pm at} \cos kt$	$\frac{s \mp a}{(s \mp a)^2 + k^2}$
8. $\sin at$	$\frac{a}{s^2 + a^2}$
9. $\cos at$	$\frac{s}{s^2 + a^2}$
10. $\sinh at$	$\frac{a}{s^2 - a^2}$
11. $\cosh at$	$\frac{s}{s^2 - a^2}$
12. $t \sin at$	$\frac{2as}{(s^2 + a^2)^2}$
13. $t \cos at$	$\frac{s^2 - a^2}{(s^2 + a^2)^2}$
14. $\sin (wt + \theta)$	$\frac{s \sin \theta + k \cos \theta}{s^2 + k^2}$
15. $\cos (wt + \theta)$	$\frac{s \sin \theta - k \cos \theta}{s^2 + k^2}$

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Properties of Laplace Transform**1. Linearity** Excel Review Center

$f(t)$	$F(S)$
$af_1(t) + bf_2(t)$	$aF_1(s) + bF_2(s)$

2. Scaling

$f(t)$	$F(S)$
$f(at), a > 0$	$\frac{1}{ a } F\left(\frac{s}{a}\right)$

3. Time-domain shift

$f(t)$	$F(S)$
$f(t-a)$	$e^{-as} F(s)$

4. s-Domain shift

$f(t)$	$F(S)$
$e^{s_1 t} f(t)$	$F(s-s_1)$

Time Domain differentiation

$f(t)$	$F(S)$
$\frac{d}{dt} f(t)$	$sF(s) - x(0)$

6. s-Domain differentiation

$f(t)$	$F(S)$
$-tf(t)$	$\frac{d}{ds} F(s)$

7. Integration

$f(t)$	$F(S)$
$\int_0^t f(t) dt$	$\frac{F(s)}{s}$

8. Convolution Excel Review Center

$f(t)$	$F(S)$
$f(t) * h(t)$	$F(s) * H(s)$

Inverse Laplace Transform

$$f(t) = L^{-1}\{F(s)\}$$

$$f(t) = \frac{1}{j2\pi} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds$$

Fourier Transforms

Fourier Transform, named after the French mathematician **Jean Baptiste Fourier** (1768-1830), is a function derived from a given function by representing it by a series of sinusoidal functions. It is a very useful tool in describing periodic and aperiodic signals.

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$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

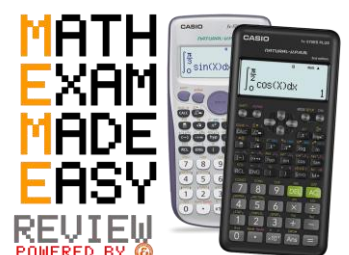
$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

The Fourier transform can be expressed by:

$$F(\omega) = F\{f(t)\}$$

Fourier Transform of some elementary functions:

$f(t)$	$F(\omega)$
1. 1	$2\pi\delta(\omega)$
2. $u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
3. $\delta(t)$	1
4. $\delta(t-t_0)$	$e^{-j\omega t_0}$
5. $e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
6. $\cos(\omega_0 t)$	$\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
7. $\sin(\omega_0 t)$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
8. $\cos(\omega_0 t) u(t)$	$\frac{\pi}{2}[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)] + \frac{j\omega}{\omega_0^2 - \omega^2}$
9. $\sin(\omega_0 t) u(t)$	$j\frac{\pi}{2}[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)] + \frac{j\omega}{\omega_0^2 + \omega^2}$
10. $e^{-a t } u(t)$	$\frac{1}{a + j\omega}$
11. $te^{-a t } u(t)$	$\left(\frac{1}{a + j\omega}\right)^2$
12. $e^{-a t }, a > 0$	$\frac{2a}{a^2 + \omega^2}$
13. $te^{-a t }$	$\frac{j4a\omega}{a^2 + \omega^2}$



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